

课后习题(6) — 语义分析和中间代码生成

于梦洁
3021244141

1. 解: (1) 其中优先级 $\text{not} > \text{and} > \text{or}$

逆波兰式: $AB \text{ or } CD \text{ not } E \text{ not } \text{and } \text{or } \text{and } F \text{ or}$

(2) 逆波兰式: $a - b c c d e \mid - \wedge * + c - d + -$

(3) 逆波兰式: $abcde \mid + * + 0 == \text{if } s a 3 e + \wedge := \text{then } s b a d \wedge \wedge := \text{else}$

2. 解: $-(a+b) * C^{\wedge}(a*d+e) - (a+b-c)$

四元式:

	op	arg1	arg2	result
(1)	+	a	b	T ₁
(2)	uminus	T ₁		T ₂
(3)	*	a	d	T ₃
(4)	+	T ₃	e	T ₄
(5)	$\wedge$	c	T ₄	T ₅
(6)	*	T ₂	T ₅	T ₆
(7)	+	a	b	T ₇
(8)	-	T ₇	c	T ₈
(9)	-	T ₆	T ₈	T ₉

间接三元式:

	op	arg1	arg2
(1)	+	a	b
(2)	uminus	(1)	
(3)	*	a	d
(4)	+	(3)	e
(5)	$\wedge$	c	(4)
(6)	*	(2)	(5)
(7)	+	(1)	c
(8)	-	(6)	(7)

间接代码表:

(1) (2) (3) (4) (5) (6) (7) (8)

三元式:

	op	arg1	arg2
(1)	+	a	b
(2)	uminus	(1)	
(3)	*	a	d
(4)	+	(3)	e
(5)	^	c	(4)
(6)	*	(2)	(5)
(7)	+	a	b
(8)	-	(7)	c
(9)	-	(6)	(8)